Engineering Mechanics 1 · Unit_02 · UNIT 2

DRAFT — UE26ME142A · Unit_02 · UNIT 2 · UE26ME142A Unit 2 Class 24 AV Summary

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A Unit 2 Class 24 AV Summary
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

UE26ME142A Unit 2 Class 24 AV Summary

1 UE26ME142A Unit 2 Class 24 AV Summary

Study focus: UE26ME142A_Unit 2_Class 24_AV Summary. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000242 · EV0000243

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

My name is Dr.

Key details

- Sv Venkitish from Civil Engineering under the Department of Science and Humanities, PS University.
- This is class number 33.
- In the previous classes we had discussed about chapter four, which is on trusses under unit two.
- We had solved a few problems, so we'll be continuing solving a few more problems in today's class.
- The statement of the problem goes like this determine the forces in members A B, B C and B D.
- of the loaded truss.
- Why it is not symmetric?
- Because we have a horizontal force here at D.
- And if you take the symmetric axis as B D.
- Of course it is symmetric with respect to geometry because you have the same number of members to the left as well as the two right of this line B.
- However, the loading you have only at T which is acting towards the right, we don't have a loading towards the left.
- Supports that we have here is an hip support and a roller support.
- So since there is one vertical and one horizontal, we will also have a horizontal reaction because we have an horizontal force there, yeah.

- Next, we have to determine the forces in few of the members, whatever is asked according to the problem.
- We will first have to determine the support reactions because we cannot start with any of the joints, which has got only two unknowns, where we can determine the forces.
- So we need to first find out the support reactions.
- Which is also equal to L by 2.
- Then if you take the algebraic sum of all the forces along X direction, we get the reaction at A, which will be acting towards the left, which is equal to L.
- Next we can take any of the joint, either joint C or A.
- The force in the current member will be or should be taken to be same as the force in the member where you join that two points or the joints of this member.
- Find out what is the force that is going to be acting along that line C D.
- The same will be the force that is going to be acting in the current member C D as well.
- So the angle made by DAB will be 45 degrees because this is also R, this is also R so I can resolve the force in the member A D along the vertical and horizontal and use the equations of equilibrium.
- Then you take the force in the sorry you will have the.
- I am sorry about that.
- First you resolve this force along vertical and horizontal.
- You take the algebraic sum of all the forces along vertical.
- Sine 45 to be equal to the reaction that's going to be acting here, which will be acting downwards, which is equal to L by 2.
- Then you take all the forces along the x-axis, so we get the force in the member AB.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 05:00–09:59

plus the force, the component of the force in the member F A D sine forty five that will be this one equal to the force that is L acting along the horizontal reaction at A.

Key details

- Then we take this joint to B.
- The force in the member B D will be zero because there is no force acting at B.
- Then the force in the member B C will be equal to the force in the member AB itself.
- Then you take joint D.
- Then we will get from joint D the force in the member B C.
- So we get the force in the member A D because here we have determined the force from joint C, then B, then we have taken D.
- But here it is joint C, then joint B and joint D.
- However you get these values.
- You get it as negative minus 0.707 L that is in this member.
- Since it is negative, it is going to be compressive.
- Then force in the member B C, we will get it as 0.5 L, which is positive, so it is tension.

- Force in the member B D will be 0 because there is no external force acting at B.
- Then force in the member AB will be the same as the force in the member B C, which is 0.5 L, which is tension.
- Force is the member, A D is also tensile, so that we will get.
- Truss would try to rotate in the clockwise direction.
- So, since this is going to rotate, then the member AD will get elongated because the force or the reaction acting here is acting downwards, and this also will be acting towards the left.
- Then when this tries to rotate along this, this member and this member would try to go down because here these two will not move, only this can move.
- Come down because of that, your AD has to elongate and AB also has to elongate.
- Both of them has to be intentional and that's what we have got here.
- I hope this you have understood.
- Of course I have explained how to determine the forces in all the members and as well as how to determine the reactions at A and C.
- And I have also told you why this truss is not a symmetrical truss.
- Moving on, I have chosen this problem, which is 4 bar 21.
- The statement of the problem goes like this.
- The signboard truss is designed to support an horizontal wind load of 4 kilometer.
- A separate analysis shows that 5 by 8 of this force is transmitted to the center connection at sea.
- And the rest is equally divided between B and B.
- Calculate the forces in member BE and BC.
- From the statement, what we can understand is there is a truss which is designed for supporting a sign mode.
- The total force that is going to be acting on the signboard because of the wind load.
- They are told what will be the portion of this total load which will be acting at each of the joints at that is at P, C and D.

Example or application from the lecture

- Wind load will be acting on an area, but they are given what is the total load that is going to be acting on this signboard as well.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

Core explanation · 10:00–14:59

And for that, there is a separate analysis that they have done, which is not included in the in this in this syllabus.

Key details

- He says that five by eight of the total load will be acting at C.
- The four that is going to be acting will be three by eight into half of that into four will be acting at D and the same load will also be acting at B.
- And also what is asked is to determine the force in the member PE and BC.

- That is force in the member BE and BC.
- Get the value of in these two members, then take joint C, get the value in these two members, then take joint B, sorry, joint E, and then get these two members.
- From this we will be able to get the value of B and B C.
- If you take the entire truss as it is, it is supported at A by a pin connection, at G also it is supported by a pin connection.
- Here also will have two reactions, one vertical and one horizontal.
- So which means that it is an indeterminate structure, we require one more equation.
- However, to determine the forces in the member BE and BC, we don't have to.
- Find out the value of the support reaction first thing.
- The second thing is.
- Since we are he has asked us to find out only the forces in B and B C, we have to solve only the joint B, joint C, and joint E only.
- We don't have to worry about solving other joints.
- How the forces is going to be acting here.
- They have explained how to get the forces in AB obviously and sigma m b equal to zero by using alternate equations of equilibrium.
- Then use the.
- Equation sigma m d equal to 0.
- Now this method that they have used is method of section.
- We will be using as I described by taking the joint D analyzing this, getting these two equations.
- or getting these value of these two, then analysing joint C, getting the value of C E and C B, and then using the joint E to get the force in the member B.
- Now the next problem that is chosen is 4 bar 23.
- Which says that if snow load transfers the forces shown to the upper joint of a prat roof truss, neglect any horizontal reactions at the supports and all for the forces in all the members.
- And if you observe all the forces are vertically acting downwards and it is equal to one kilometer.
- And we have the reaction acting at A, which is a pin connection, we have the reaction acting at E, which is also a pin connection.
- But since there is no horizontal.

Example or application from the lecture

- Let's look at what are the forces that is acting.
- Then the remaining load, that is 3 by 8, which is remaining of the four kilometers, is to be equally shared at D and C is what is given.
- Let us not worry about this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 15:00–19:59

Force that is acting, we can say that the horizontal reaction at A and E will be equal to 0.

Key details

- And since the trust is symmetrical with respect to geometry as well as loading as well as the support reactions, we can say that the vertical reaction at A and E will be equal, and which will be equal to 2.5 kilonuton, which is nothing but 1 plus 1 plus 1 plus 1 plus 1, which is 5 kilonuton divided by 2.
- We get the value of alpha standing inverse of 2 by 4, and that's what is indicated here.
- Using that and considering the joint A, we get the force in the member AB under AH.
- By using the equations of equilibrium, we can get the value of AB and AH.
- Next, to solve this, first we will solve joint A as I've explained just now, then go to joint B.
- There are only two members, we can get the value in BG as well as B H.
- Then we can take the joint H, we can get the value in H G and H C.
- Then we can take the joint G.
- Where we can easily identify that the force in the member CG is zero because there is no vertical external force that is acting at G.
- Since these two is going to be equal, the force in the member CG will be zero.
- Only these four joints is what we have to solve.
- Next, you take joint B.
- What are the forces acting?
- One is external force that is acting because of the snow load one kilon.
- Then we have BH acting.
- Then we have the force in this which we have determined as 3.35 kilon tons, then the force in BC.
- We can find out what is the force in B C, which we will get as 3.35 kilonewtons.
- Then you take sigma f y equal to 0, we get the force in the member bh as 1 kilowton.
- Both of them are acting at the same angle, alpha, so these two has to be equal, so we get as three point three five.
- When I resolve this along vertical and resolve this as along vertical, they both will cancel, so we get VG as one kilometer straight away.
- Then you take the joint h and then determine what is CH and CG, resolve CH along vertical and horizontal, take sigma y equal to 0, we get ch as 1 point four one four.
- Then sigma f x equal to 0, if you take I get the force in the member gh.
- Then Cd is equal to B C, which is equal to 3.35 kilonewton, which is also compressive.
- Then EF is equal to.
- A H which is equal to 3 kilonut time which is tensile, then Bf is equal to BH which is equal to 1 kilonut time which is compressive.
- Cg is equal to CH which is equal to 1.414 kilonut time which is tension.
- Fg is equal to G H, which is equal to two kilone, which is tension, and of course CG force in the member CG is equal to zero.
- Now because of symmetry.
- We are able to get the forces in all the members by solving only three joints, and of course the joint G, if you take you get the force in the member CG.

- Because we have the same forces acting to the left of the symmetrical axis and right of the symmetrical axis.
- I am Dr.
- Steven Tesh from Civil Engineering, the Department of Science and Humanities, PS University.

Example or application from the lecture

- And that is what is given here.
- Next, consider the angle made by the line B C with the horizontal, that is angle CAG to be equal to alpha.
- Let's meet in the next class, that is class number thirty-four, and discuss more about.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
UE26ME142A_Unit 2_Class 24_AV Summary transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
UE26ME142A_Unit 2_Class 24_AV Summary transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
UE26ME142A_Unit 2_Class 24_AV Summary transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_02/AV_Summaries/024_UE26ME142A_Unit_2_Class_24_AV_Summary__ddeae52719c2a16c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 014_UE26ME142A_Unit_2_Class_14_Lecture_PPT__68b1ea1b872acf84.slides.md have no readable `.slides.md` text; their original page images are included.
2. 4 rendered pages in 015_UE26ME142A_Unit_2_Class_15_Lecture_PPT__ad0881da453d8078.slides.md have no readable `.slides.md` text; their original page images are included.
3. 2 rendered pages in 016_UE26ME142A_Unit_2_Class_16_Lecture_PPT__4b521e78b51c81b7.slides.md have no readable `.slides.md` text; their original page images are included.
4. 4 rendered pages in 017_UE26ME142A_Unit_2_Class_17_Lecture_PPT__036572c14ae14d94.slides.md have no readable `.slides.md` text; their original page images are included.
5. 3 rendered pages in 018_UE26ME142A_Unit_2_Class_18_Lecture_PPT__a852acc6b8941200.slides.md have no readable `.slides.md` text; their original page images are included.
6. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.0_Lecture_PPT__085bbdaae12394ff.slides.md have no readable `.slides.md` text; their original page images are included.
7. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.1_Lecture_PPT__14bcc58873e77b45.slides.md have no readable `.slides.md` text; their original page images are included.
8. 4 rendered pages in 019_UE26ME142A_Unit_2_Class_19.2_Lecture_PPT__444fedd8753b84b9.slides.md have no readable `.slides.md` text; their original page images are included.

9. 3 rendered pages in 019_UE26ME142A_Unit_2_Class_19.3_Lecture_PPT__7981db4bc773065f.slides.md have no readable `.slides.md` text; their original page images are included.
10. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.4_Lecture_PPT__54dd570fb45dd727.slides.md have no readable `.slides.md` text; their original page images are included.
11. 2 rendered pages in 019_UE26ME142A_Unit_2_Class_19.5_Lecture_PPT__c337d52a52027bf0.slides.md have no readable `.slides.md` text; their original page images are included.
12. 3 rendered pages in 021_UE26ME142A_Unit_2_Class_21_Lecture_PPT__66c97ce40e0b782a.slides.md have no readable `.slides.md` text; their original page images are included.
13. 2 rendered pages in 022_UE26ME142A_Unit_2_Class_22_Lecture_PPT__b1114b6d964d54c7.slides.md have no readable `.slides.md` text; their original page images are included.
14. 3 rendered pages in 023_UE26ME142A_Unit_2_Class_23_Lecture_PPT__433882c2bbb0b6a0.slides.md have no readable `.slides.md` text; their original page images are included.
15. 2 rendered pages in 024_UE26ME142A_Unit_2_Class_24_Lecture_PPT__c0fa0555c4f1df8b.slides.md have no readable `.slides.md` text; their original page images are included.
16. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
17. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
18. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
19. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
20. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00058	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00059	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00060	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00061	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

C Revision and retrieval practice